

PUBLIC WORKS DEPARTMENT (PWD)

Ministry of Housing and Public Works, Government of Bangladesh

API INTEGRATION DEMAND DOCUMENT

PWD Asset Management Database — BIMH Establishment Data Lookup

Document Control

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Prepared For	Battery Low Interactive Limited (BIMH Developer)
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1. Purpose

The Public Works Department (PWD) is developing a web-based PWD Asset Management Database to maintain a comprehensive register of all government structures managed by PWD. A key requirement of this system is the ability to link each asset record to its official establishment entry in the BIMH platform, operated by Battery Low Interactive Limited.

This document formally specifies the API integration that PWD requires from the BIMH service provider. The integration enables a user of the PWD Asset Management Database to enter a BIMH Establishment ID and automatically retrieve approved establishment attributes from the BIMH system, eliminating manual data re-entry and ensuring correctness of structural reference data.

2. Scope

This document covers the following:

- Authentication — obtaining a bearer token from the BIMH API.
- Establishment Data Lookup — retrieving the establishment profile for a given BIMH ID.

The following are explicitly out of scope:

- Budget data APIs.
- Notification or callback APIs (unless required as part of the authentication or lookup flow).
- Any write/push operation from PWD systems to BIMH.

3. Operating Context

The PWD Asset Management Database will introduce a dedicated BIMH field type for asset records. When a user enters or uploads a BIMH Establishment ID, the system will call the BIMH API server-side, retrieve the establishment profile, display the fetched attributes as read-only fields in the user interface, and store the values locally in the application database.

The key design principles are:

- Ordinary table browsing and asset display must not depend on live BIMH API availability; only the initial fetch relies on the API.
- BIMH attributes are system-managed and read-only in the consuming application; only the BIMH ID is user-editable.
- Both manual asset entry and Excel bulk upload are supported; the fetch is triggered explicitly in manual mode and automatically during bulk audit review.

4. Authentication

4.1 Token Endpoint

Based on the reference documentation provided by the BIMH team, PWD understands the token mechanism as follows:

#	Parameter	Type	Description
1	Method	–	POST
2	URL	–	{base-url}/api/token
3	Content-Type	Header	application/x-www-form-urlencoded
4	Authorization	Header	Basic {hash_value_key}

4.2 Token Request Body

#	Parameter	Type	Description
1	grant_type	String	Authentication grant type (e.g. password)
2	username	String	API username issued to PWD by BIMH provider
3	password	String	Corresponding API password

4.3 Expected Token Response Fields

#	Parameter	Type	Description
1	access_token	String	Bearer token to include in all subsequent API calls
2	token_type	String	Expected value: bearer
3	expires_in	Integer	Token lifetime in seconds
4	refresh_token	String	Token for renewing the session (if applicable)
5	responsecode	String	0 = success; other codes indicate failure

Please confirm: (a) the exact value and construction rule for the Basic authorization hash header, (b) whether a refresh token is issued and how to use it, and (c) whether the token can be cached and reused until expiry or must be re-requested per session.

5. Establishment Data Lookup API

5.1 Endpoint Summary

#	Parameter	Type	Description
1	Method	–	GET (or as confirmed by provider)
2	URL	–	{base-url}/api/PWD_BIMH/GetPWDEstablishmentAssetData
3	Authorization	Header	Bearer {access_token}
4	Content-Type	Header	application/json

5.2 Request Parameters

PWD understands the request to require the following parameters. Provider must confirm the exact parameter name for the BIMH Establishment ID.

#	Parameter	Type	Description
1	requestId	String (32)	A unique 32-character request identifier generated by PWD for each call
2	ministryCode	String (3)	3-digit ministry code for PWD (to be confirmed by provider)
3	[BIMH ID param]	String	Exact parameter name to be confirmed — carries the BIMH Establishment ID entered by the user

5.3 Sample Request

```
GET {base-url}/api/PWD_BIMH/GetPWDEstablishmentAssetData
Authorization: Bearer <access_token>
Content-Type: application/json
```

```
{
  "requestId"      : "abcd1234efgh5678ijkl9012mnop3456",
  "ministryCode"   : "144",
  "[bimhIdParam]" : "EST-00001"
}
```

6. Required Response Fields

The table below lists the establishment attributes that the PWD Asset Management Database requires. Fields marked Required must always be present in the response; Optional fields should be included when available.

#	Field Name	Data Type	Constraints	Description
1	establishmentId	Number (18)	Required	Unique BIMH establishment identifier
2	establishmentNameEn	String (1000)	Required	Official establishment name in English
3	concernedMinistry	String (200)	Required	Ministry responsible for the establishment
4	constructedBy	String (200)	Required	Executing agency (e.g. PWD, LGED)
5	administrativeDivision	String (100)	Required	Administrative division name
6	administrativeDistrict	String (100)	Required	Administrative district name
7	administrativeUpazila	String (100)	Required	Administrative upazila name
8	administrativeUnion	String (100)	Required	Administrative union name (nullable)
9	civilZone	String (100)	Required	PWD Civil Zone authority
10	civilCircle	String (100)	Required	PWD Civil Circle Authority
11	civilDivision	String (100)	Required	PWD Civil Division Authority
12	civilSubdivision	String (100)	Required	PWD Civil Sub-division Authority
13	mechanicalZone	String (100)	Required	PWD E/M Zone authority
14	mechanicalCircle	String (100)	Required	PWD E/M Circle Authority
15	mechanicalDivision	String (100)	Required	PWD E/M Division Authority
16	mechanicalSubdivision	String (100)	Required	PWD E/M Sub-division Authority
17	latitude	String (50)	Required	GPS latitude in decimal format
18	longitude	String (50)	Required	GPS longitude in decimal format
19	structuralDrawingID	String (50)	Required	Structural Drawing ID
20	architecturalDrawingID	String (50)	Required	Arc Drawing ID
21	yearofConstruction	Number (5)	Required	Year of construction (numeric)
22	usesOfEstablishment	String (100)	Required	Functional use (Official, Residential, etc.)
23	establishmentHeight	Number (5)	Required	Height of structure
24	aboveGroundStoreys	Number (3)	Required	Number of floors above ground
25	underGroundStoreys	Number (3)	Required	Number of floors above ground
26	plinthArea	Number (7)	Required	Footprint area of the building

27	builtUpArea	Number (7)	Required	Total built-up floor area
28	structureType	String (100)	Required	Structural frame type (e.g. RC Frame, Steel, Masonry)
29	foundationType	String (100)	Required	Foundation type (e.g. Pile, Mat, Footing)
30	foundationFor	Number (3)	Required	Number of floor the building's foundation can accomodate
31	liftNumber	Number (3)	Required	Total number of lifts installed
32	acNumber	Number (3)	Required	Total number of air-conditioning units
33	acCapacity	Number (5)	Required	Total AC cooling capacity
34	generatorNumber	Number (3)	Required	Number of backup generators
35	generatorCapacity	Number (5)	Required	Capacity of generators (e.g. 300 KVA)
36	substationNumber	Number (3)	Required	Number of electrical substations
37	substationCapacity	Number (5)	Required	Capacity of substations (e.g. 500 KVA)
38	motorNumber	Number (3)	Required	Number of water pumps / motors
39	motorCapacity	Number (5)	Required	Motor capacity (e.g. 10 HP)
40	fireDetectionSystem	String (1000)	Required	Description of fire detection devices installed
41	fireProtectionSystem	String (1000)	Required	Description of fire protection systems installed

6.1 Sample Response (Partial)

```
{
  "requestId" : "abcd1234efgh5678ijkl9012mnop3456",
  "data" : [
    {
      "establishmentId"      : "EST-00001",
      "establishmentNameEn"  : "PWD Head Office",
      "concernedMinistry"    : "Ministry of Housing and Public Works",
      "constructedBy"        : "PWD",
      "administrativeDivision" : "Dhaka",
      "administrativeDistrict" : "Dhaka",
      "administrativeUpazila"  : "Ramna",
      "latitude"              : "23.7382",
      "longitude"             : "90.3952",
      "buildingCategory"      : "A",
      "constructionCompletionDate": "2020-12-31",
      "structureType"         : "RC Frame",
      "foundationType"        : "Pile",
      "buildingCondition"     : "Good",
      "liftNumber"            : 4,
      "acNumber"              : 20
      ...
    }
  ]
}
```

7. Error Handling Requirements

The consuming application requires deterministic, machine-readable error responses for the conditions listed below.

HTTP Status	App Code	Meaning / Expected Behavior
200	0	Successfully processed — data returned
200	1	Invalid input parameter — request malformed or missing field
200	2	Token generation failed
200	3	Authentication failed — wrong username or password
200	5	Token expired — client must re-authenticate
400	—	Bad request — general malformed request
401	—	Unauthorized — bearer token absent or invalid
404	—	Establishment ID not found — no matching record in BIMH
500	—	Upstream server error — client should display connection error and allow retry
429	—	Rate limit exceeded — provider must confirm threshold and retry guidance

For an invalid or unrecognized BIMH Establishment ID, PWD expects a clear, distinguishable response (HTTP 404 or an explicit application-level code) so the consuming UI can display an appropriate status and prompt the user to correct the ID.

8. Security Requirements

- All API communication must use HTTPS / TLS. Plaintext HTTP is not acceptable.
- API credentials (username, password, hash key) must be rotatable by the provider without service interruption.
- Bearer tokens must carry a stated expiry; the consuming system will not cache a token beyond its expiry.
- Error messages must not include raw stack traces or internal system paths.
- The provider should confirm whether API requests are logged and what identifiers are retained, for joint incident troubleshooting.

9. Performance & Usage Constraints

Please confirm the following so the consuming application can be implemented safely:

- Maximum number of API requests per minute / hour (rate limit).
- Whether sequential bulk requests (one BIMH ID per row, during Excel bulk import audit) are acceptable or whether batching is preferred.
- Recommended retry strategy on timeout or 5xx errors.
- Planned maintenance windows or known downtime schedules.

10. Deliverables Requested from Provider

PWD requests the following deliverables from Battery Low Interactive Limited to finalize the integration:

#	Deliverable	Preferred Format
1	Production and sandbox base URLs	Plain text / Markdown
2	IP whitelisting requirement and VPN/network access details	Plain text
3	Exact Basic authorization hash construction rule	Markdown with example
4	Complete token request and response examples	Markdown / JSON
5	Confirmed endpoint URL and exact BIMH ID parameter name	Markdown / OpenAPI
6	Complete response field dictionary with data types	Markdown table / XLSX
7	Error code catalogue and HTTP status mapping	Markdown table
8	Sample valid and invalid BIMH IDs for testing	Plain text
9	Rate limit and retry policy documentation	Markdown / PDF
10	Change management and deprecation notice procedure	Plain text / PDF

11. Open Items Pending Provider Confirmation

#	Parameter	Type	Description
1	Base URL (production & sandbox)	–	Not yet provided
2	Exact BIMH lookup endpoint path	–	Assumed from reference doc; must be confirmed
3	Exact request parameter name for BIMH ID	–	Not confirmed
4	Basic auth hash generation rule	–	Formula not fully specified
5	Refresh token behavior	–	Not specified
6	Definitive response field list	–	Pending field dictionary
7	Rate limit and retry guidance	–	Not specified
8	Test/sandbox credentials	–	Not yet issued

12. Recommended Next Step

PWD requests that Battery Low Interactive Limited review this document and provide the deliverables listed in Section 10, preferably in Markdown or PDF format with concrete worked examples. Upon receipt, the PWD Software Team will finalize the server-side API client implementation and the field mapping configuration.

For queries or clarifications on this document, please contact the PWD Software Team.